Predicting

Flu Vaccine Uptake

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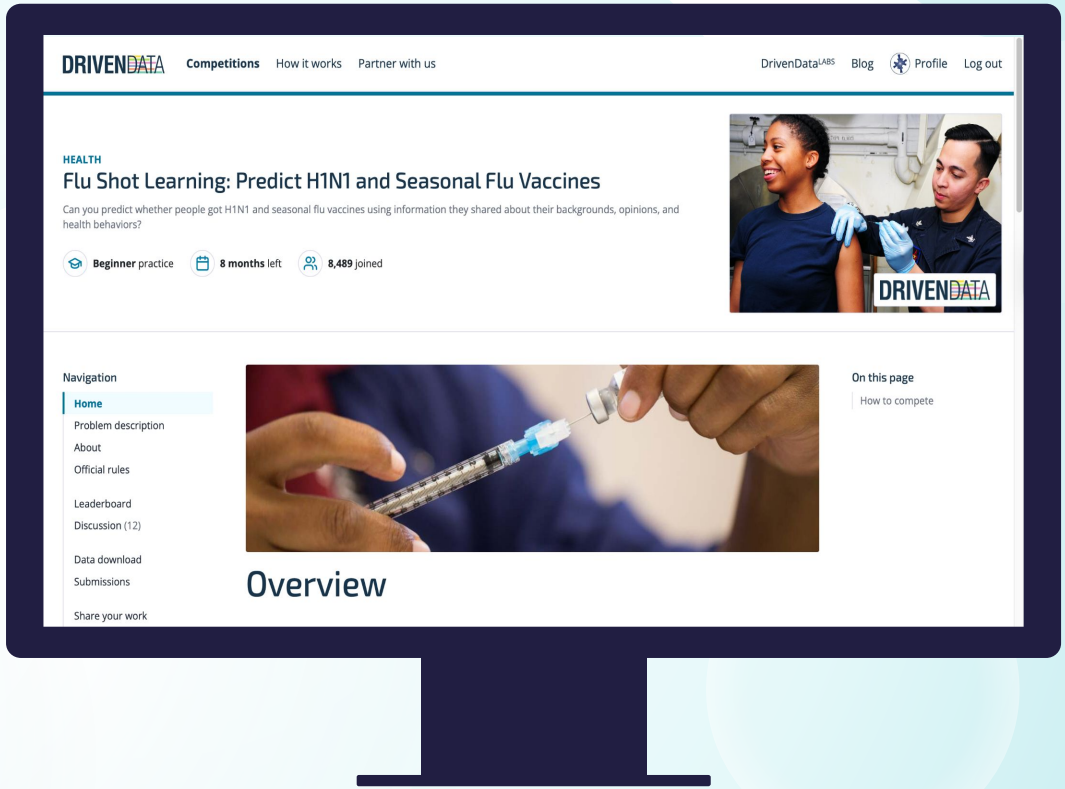
Chosen Competition

Flu Shot Learning:
Predict H1N1 and Seasonal Flu Vaccines

Data:
National 2009 H1N1 Flu Survey (NHFS)



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Business Problem

VaxCare:
Producer of pharmaceutical vaccines (seasonal flu)

Problem:
Since 2021, 10% decline in U.S. adults opting to receive flu vaccines (CDC)

Implications:
⬇ $12.5 million in VaxCare revenue
⬆ Increase in societal burden ($11.2 billion)

The Model

Goal

Use backgrounds, opinions, & health behaviors to **predict whether they received** the H1N1 and seasonal flu vaccines.

Value

Identify populations least likely to be vaccinated and target those individuals with messaging.

- Reduce marketing costs
- Support public health
- Drive efficient revenue growth

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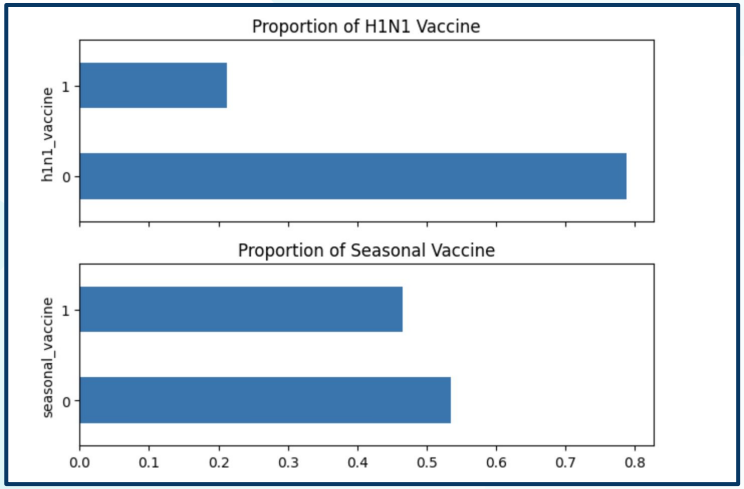
Exploratory Data Analysis and Processing



- **~20% of people received the H1N1 vaccine**
- **~50% of people received the seasonal flu vaccine**

Missing Values:

- Numerical → Median Value
- Binary/Categorical → Most Frequent Value

Feature Scaling:

- Numerical features → StandardScaler
- Mean = 0, Standard Dev. = 1

One Hot Encoding:

- Categorical variables → binary array (1 selected, 0 non-selected)

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How VaxCare is Testing the Model:

Data Splitting:

- Training 60%/ Validation 20%/ Test 20%
- Reliable model development and unbiased test
- Stratified sampling to keep vaccine outcome proportions across sets

Why VaxCare Chose ROC AUC:

- Works with unbalanced data
- Consistent model performance comparison

Why ROC AUC matters for VaxCare:

- Helps ID who is most/least likely to get vaccinated
- Better performance → better marketing targeting

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MODEL Approaches - Initial AUC

Logistic Regression

Baseline Model
Mean AUC: 0.8364

Random Forest Classifier

Bagging Ensemble
Mean AUC: 0.8331

Gradient Boosting

Boosting Ensemble
Mean AUC: 0.8417

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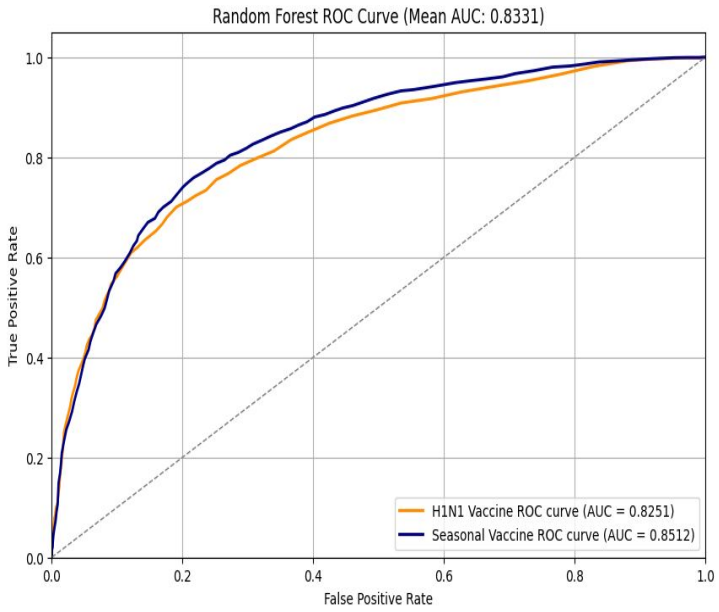
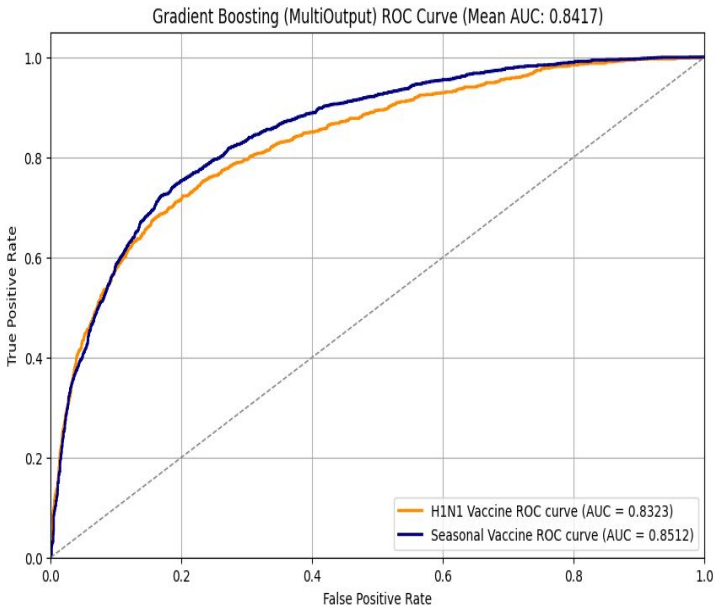
ROC Curves





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MODEL Approaches - Tuned AUC using GridSearchCV

Random Forest Classifier

Tuned Bagging Ensemble
Mean AUC: 0.8444

Gradient Boosting

Tuned Boosting Ensemble
Mean AUC: 0.8523

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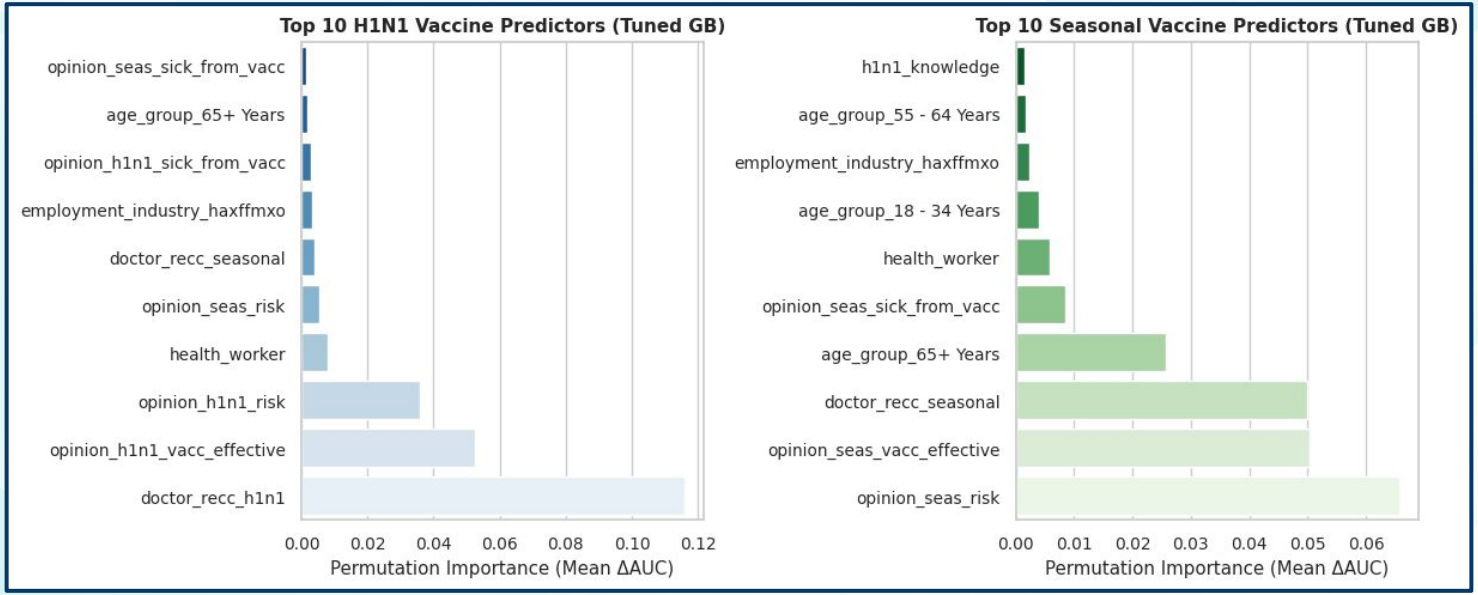
Results: Top Contributors

- Physician Recommendations
- Perception of Risk



Limitations

Potential Bias in Data

- Model is based on 2009/2010
- Behaviors could have changed since **H1N1 pandemic** and inc. in **social media information**

Generalization

- Can't 100% generalize
- Opinions may have changed as they **aged** and as **medical field evolves**

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Next Steps

A/B Testing for Messaging

- Highlight risks and physician recommendations

Collaboration with Public Health Organizations

- Well-known, trusted groups to provide support

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Thank You

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